

Product Technical Reference

SpatialTech Industries
Hardware Engineering Division
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INTERNAL & PARTNER USE

This document provides component-level technical specifications for SpatialTech's current production hardware portfolio. All values reflect shipping configurations as of firmware releases through August 2025. Specifications are subject to change with future firmware and hardware revisions. Thermal and power figures measured at 25°C ambient unless otherwise noted.

Horizon X1 — Mixed-Reality Headset

SKU Prefix: HX1 | MSRP: \$499.99

Standalone 6DoF mixed-reality headset with dual micro-OLED displays, onboard Snapdragon XR2+ Gen 2 processing, and RGB passthrough. Supports tethered PC mode via USB-C DisplayPort Alt Mode for compute-intensive rendering. Designed for enterprise simulation, architectural visualization, and developer applications.

DISPLAY & OPTICS		TRACKING & INPUT	
Display Type	Dual micro-OLED	Tracking Method	Inside-out, 6 camera array
Resolution (per eye)	3840 × 3840 (8K combined)	Hand Tracking	Yes (v2.4 firmware+)
Refresh Rate	72 / 90 / 120 Hz	Controller Type	SpatialTouch CT-1 (compact)
Field of View	110° horizontal, 96° vertical	Controller Battery	AA × 1, ~40 hr
Lens Type	Pancake (3-element)		
IPD Range	58–72 mm (motorized)		
Passthrough	Stereo RGB, 18 PPD		
COMPUTE & CONNECTIVITY		PHYSICAL & POWER	
Processor	Snapdragon XR2+ Gen 2	Weight (w/ strap)	586 g
RAM / Storage	12 GB / 256 GB	Battery	5,300 mAh Li-Po
Wi-Fi	Wi-Fi 6E (802.11ax)	Battery Life	2.1 hr (standalone) / 3.4 hr (tethered)
Tethered Mode	USB-C DP Alt, 5 m max	Charging	USB-C PD, 30W
Enterprise Mgmt	MDM via SpatialHub Console	Face Interface	Magnetic, hypoallergenic silicone
		Audio	Integrated near-ear speakers, 2W
		Microphone	3× MEMS array, noise canceling

Note: Hand tracking v2.4 requires firmware 4.0.8+. Controller ergonomics redesigned from Horizon S3; grip angle reduced 12° per user testing. Standalone battery life measured at 90 Hz, 50% brightness, Wi-Fi active. Included soft carrying case rated for transport only — not ruggedized.

HapticGlove V2 — Force-Feedback Input Glove

SKU Prefix: HG2 | MSRP: \$129.99

Per-finger force-feedback glove with 5-point linear resonant actuator (LRA) array and IMU-based finger tracking. Designed for simulation training, haptic prototyping, and interactive installations. Ships as single unit (L or R sold separately); compatible with Horizon X1, Horizon S3, and third-party OpenXR headsets.

HAPTICS & TRACKING

Actuator Type	LRA × 5 (per finger)
Force Range	0.2–3.8 N per actuator
Frequency Response	50–500 Hz
Finger Tracking	IMU × 5 + flex sensor × 5
Tracking Accuracy	±2.1 mm (calibrated)
Tracking Drift	<0.8 mm/hr (typ.)
Recalibration	Recommended every 60 min

PHYSICAL

Weight (per glove)	148 g (size M)
Sizes Available	S, M, L
Hand Circ. — S	165–178 mm
Hand Circ. — M	179–203 mm
Hand Circ. — L	204–229 mm
Material	Lycra/neoprene blend
IP Rating	IP20 (no moisture protection)

SOFTWARE

SDK	Unity, Unreal, OpenXR
Calibration Tool	Windows 10+ only
Firmware	v2.5.3 (current ship)

CONNECTIVITY

Wireless	Bluetooth 5.0 Classic
Antenna	Integrated PCB, 2 dBi
Range	8 m (open), 3–5 m (typical)
Latency	12 ms (typ.), 18 ms (peak)
USB Fallback	Micro-USB, wired mode

POWER

Battery	850 mAh Li-Po
Battery Life	6.2 hr (continuous haptics)
Charging	Charging case (Micro-USB)
Charge Time	1.5 hr (0–100%)
Motor Thermal Limit	65°C (auto-shutoff)

Note: Bluetooth 5.0 Classic selected for backward compatibility with Horizon S3; BLE upgrade planned for V3. Antenna gain is limited by glove form factor — connection stability degrades in RF-congested environments or beyond 5 m. Tracking drift spec assumes ambient temperature 20–28°C; performance may vary outside this range. Sizing based on hand circumference measured at metacarpals; no XS or XL sizes currently in production. Charging case shell is ABS plastic, 3 mm wall thickness.

OptiTrack Pro — Eye-Tracking Peripheral

SKU Prefix: OTP | MSRP: \$349.99

Clip-on binocular eye-tracking module for VR/AR headsets. Uses dual 200 Hz IR cameras with corneal reflection for gaze estimation and pupillometry. Designed for research, clinical assessment, UX analytics, and adaptive rendering. Outputs via proprietary SpatialEye format with CSV export.

EYE TRACKING

Camera	Dual IR, 200 Hz (native)
Gaze Accuracy	0.5° visual angle
Gaze Precision	0.08° RMS
Pupil Dilation	Yes (absolute diameter, mm)
Sampling — Standard	120 Hz (all features)
Sampling — Lab Mode	240 Hz (gaze only, no pupil)
Calibration	5-point, ~8 seconds

DATA & SOFTWARE

Output Format	SpatialEye (.seye), CSV
API	REST (local), Unity plugin
Heatmap Generation	Built-in (SpatialEye Studio)
JSON / Direct API	Not supported (roadmap H1 2026)
Companion App	SpatialEye Studio v3.8
UI Framework	Qt 5 (legacy)

COMPATIBILITY

Headset Mount	Universal clip (52–78 mm)
Verified Headsets	Horizon X1, S3, Meta Quest 3
Connection	USB-C (data + power)

PHYSICAL & POWER

Weight	34 g
Battery	920 mAh Li-Po (internal)
Battery Life	3.8 hr (standard), 2.1 hr (lab)
Battery Indicator	3-segment LED (30/60/100%)
Charging	USB-C, 10W
Firmware	v1.14.2 (current ship)

Note: Lab Mode (240 Hz) disables pupillometry, blink detection, and vergence tracking to maximize sampling rate. Standard Mode (120 Hz) enables all features. Battery indicator uses a 3-threshold system due to hardware constraints — intermediate charge states are estimated and may not reflect actual remaining capacity linearly. SpatialEye Studio UI is scheduled for a full rewrite targeting Qt 6 / modern UI framework in 2026. Current CSV export limited to gaze coordinates and timestamps; pupil data requires .seye format.

SenseField 360 — Full-Room Spatial Tracking System

SKU Prefix: SF3 | MSRP: \$749.99

Multi-sensor spatial tracking array for room-scale and multi-user motion capture. Each unit contains a 9-axis IMU fused with dual wide-angle optical sensors for sub-millimeter positional tracking. Designed for clinical motion analysis, film/game production mocap, and large-scale VR installations. Minimum 2-unit deployment; supports up to 16 units per tracked volume.

TRACKING		MULTI-UNIT SYNC	
Positional Accuracy	±0.3 mm (X/Y), ±0.7 mm (Z)	Sync Protocol	SpatialSync (proprietary)
Tracking Volume	Up to 12 m × 12 m (4-unit)	Calibration	Auto (firmware v3.2+)
Simultaneous Subjects	Up to 8 (full skeleton)	Max Units	16 per SpatialHub host
Update Rate	360 Hz		
Latency	<3 ms (sensor to host)		
IMU	9-axis (accel/gyro/mag)		
Z-Axis Accelerometer	MEMS, ±16g, 14-bit ADC		
X/Y-Axis Accelerometer	MEMS, ±16g, 16-bit ADC		
Optical Sensors	2× global shutter, 1280×800		
CONNECTIVITY		PHYSICAL & POWER	
Host Connection	Ethernet (PoE+) or USB-C	Dimensions (per unit)	142 × 98 × 67 mm
API	Unity, Unreal, C++ SDK	Weight (per unit)	410 g
Data Output	SpatialMocap (.smcap), FBX	Mounting	¼-20 thread + wall bracket
		Power Draw	18W (typical), 24W (peak)
		Thermal — Ambient	Operating: 10–35°C
		Thermal — Surface	Up to 58°C under sustained load
		Cooling	Passive (aluminum heatsink)
		IR Illumination	850 nm flood, Class 1 eyesafe
ENVIRONMENTAL			
IP Rating	IP30		
Light Sensitivity	<500 lux recommended		
Firmware	v3.2.1 (current ship)		

Note: Z-axis positional accuracy is lower than X/Y due to the 14-bit ADC on the Z-axis accelerometer (vs. 16-bit on X/Y). This is a known limitation of the current IMU package (STMicro LSM6DSO32) and is partially compensated by optical sensor fusion. Accuracy degrades further in volumes exceeding 10 m on the Z-axis. Surface temperature of 58°C is within operating spec but may be uncomfortable to touch; ensure adequate clearance from occupants and flammable materials. Optical tracking performance degrades above 500 lux due to IR interference from ambient light sources, particularly direct sunlight. Mounting brackets shipped separately since March 2025 due to supplier change — allow 5–7 business days additional lead time.

¹ All specifications reflect production units as of September 2025. Pre-production and engineering samples may differ. Specifications subject to change without notice.

² Firmware version references indicate minimum versions for stated capabilities. Older firmware may not support all listed features.

³ SpatialTech Industries makes no warranty regarding fitness for medical diagnostic use. Clinical and research applications should validate independently per applicable regulatory requirements (FDA, CE, etc.).

■ HapticGlove V3 specifications are not included in this document. Contact Product Engineering for pre-release technical briefs under NDA.